

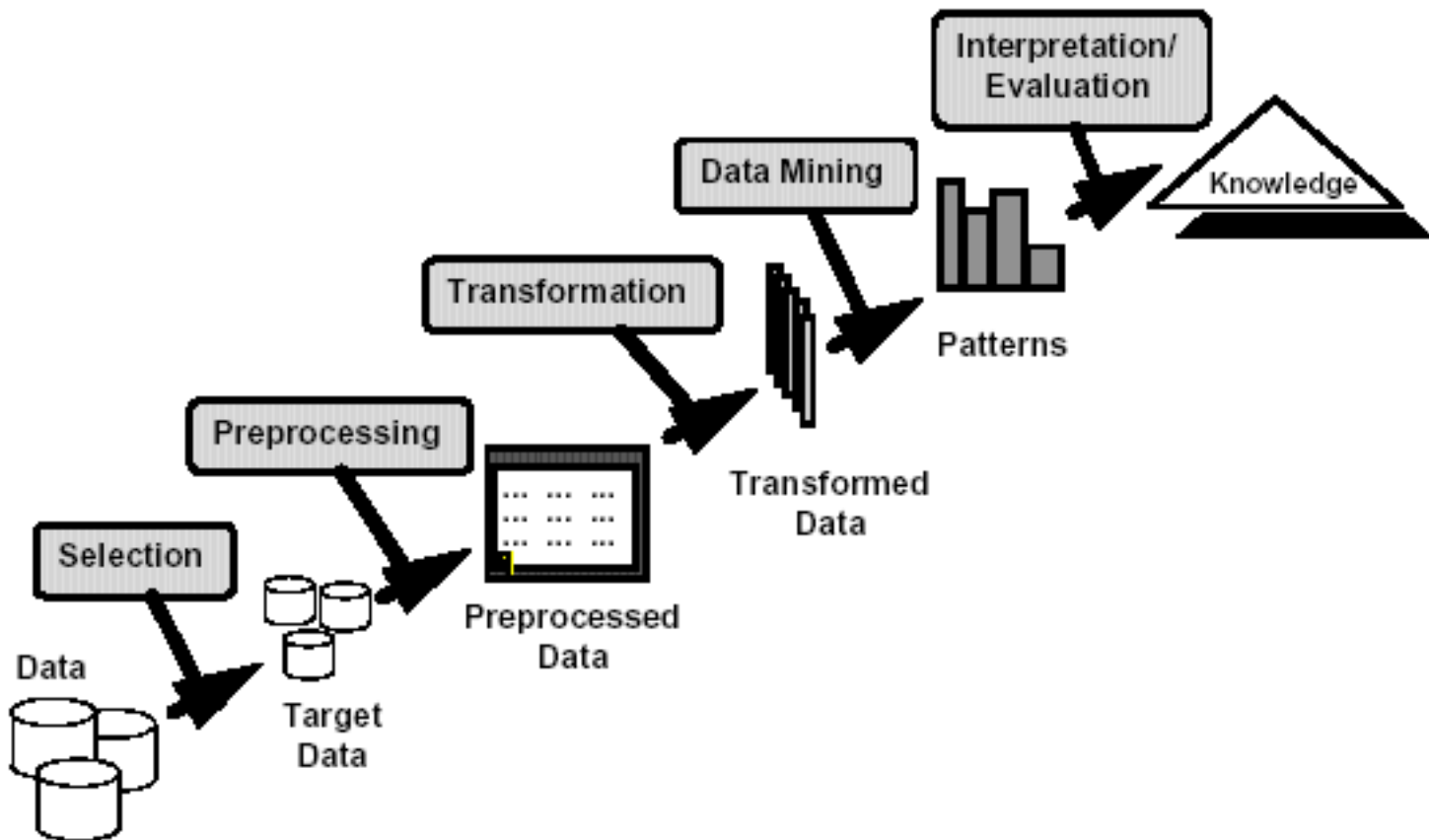
Data Mining: Summarization

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What is Data Mining?



Data Mining Tasks

- Prediction Methods
 - Use some variables to predict unknown or future values of other variables.
- Description Methods
 - Find human-interpretable patterns that describe the data.

DM: Prerequisites

- Probability, Event and Random Variable
- Maximum Likelihood Estimation (MLE)
- Maximum A Posterior Estimation (MAP)
- Convex Function and Convex Optimization
- Gradient Descent
- Simple Matrix Algebra

MLE vs. MAP

- Maximum Likelihood estimation (MLE)

Choose value that maximizes the probability of observed data

$$\hat{\theta}_{MLE} = \arg \max_{\theta} P(D|\theta)$$

- Maximum *a posteriori* (MAP) estimation

Choose value that is most probable given observed data and prior belief

$$\begin{aligned}\hat{\theta}_{MAP} &= \arg \max_{\theta} P(\theta|D) \\ &= \arg \max_{\theta} P(D|\theta)P(\theta)\end{aligned}$$

When is MAP same as MLE?

Gradient Descent Algorithm

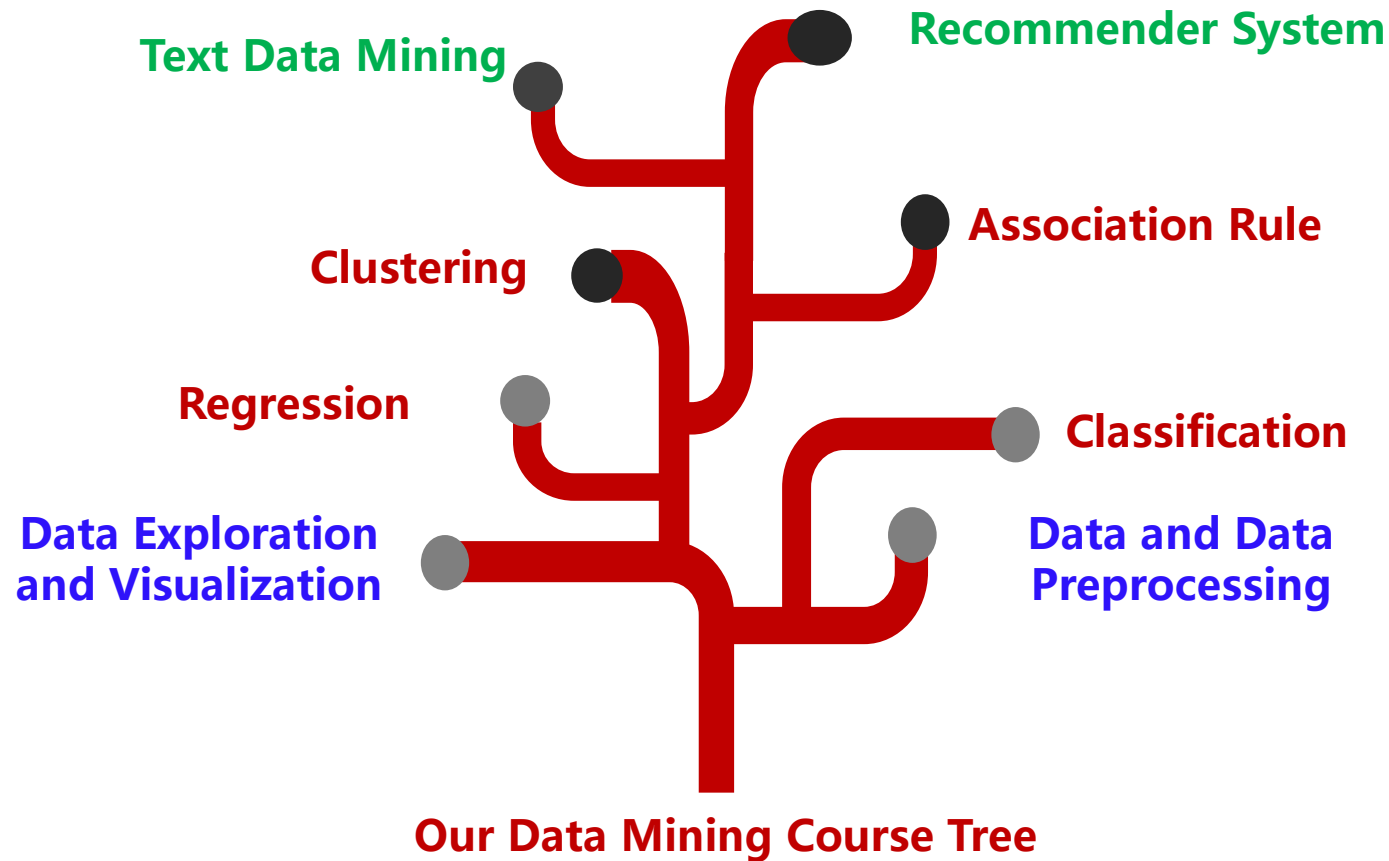
- Minimize a convex function: $\min f(x)$
- Choose initial $x^0 \in R^p$
- Iterate over $t = 0, 1, 2, \dots$ until some stopping criteria

$$x^{t+1} = x^t - \alpha_t \nabla f(x^t)$$

Stop when some criterion met:

e.g. fixed # iterations, or Gradient is within a predefined threshold

Topics Covered



DM and ML

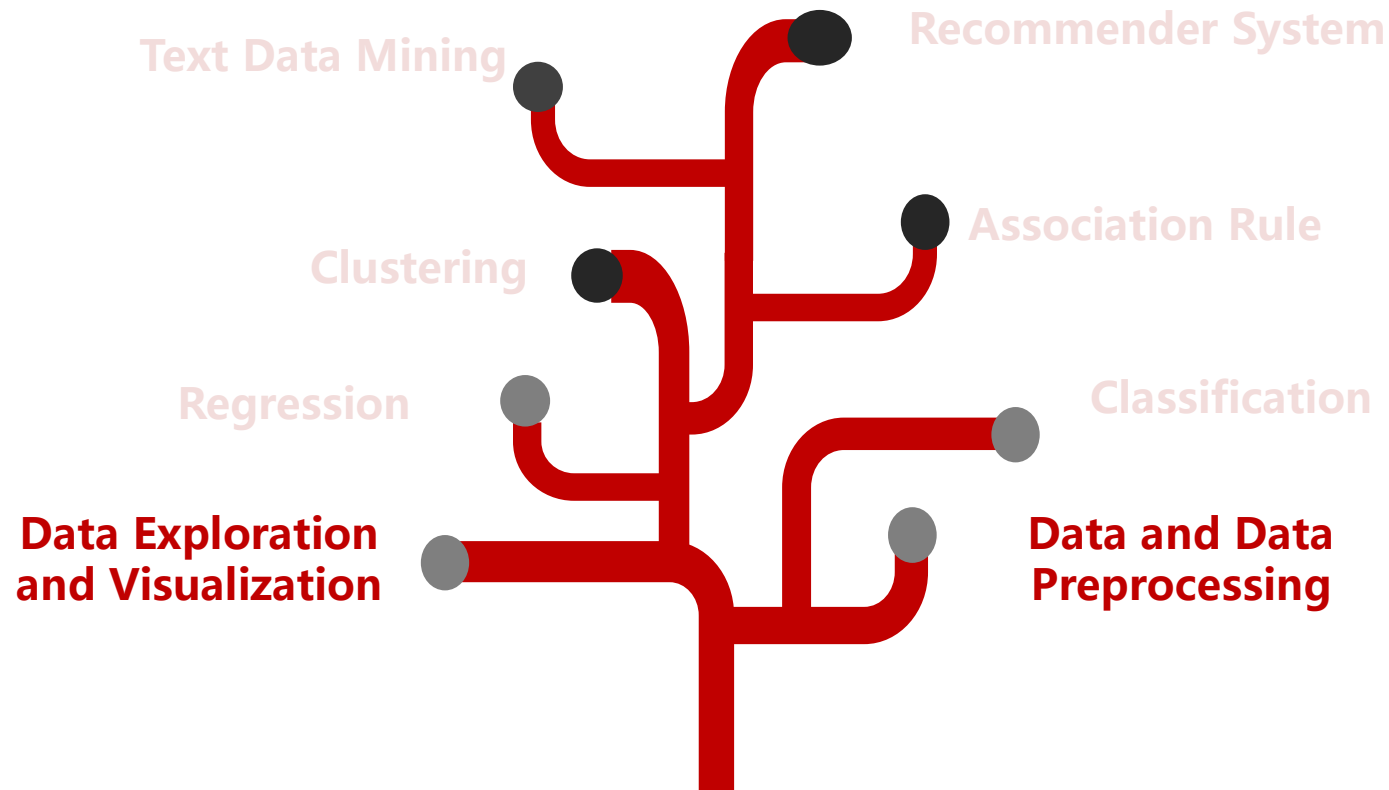
Supervised

- Classification
- Regression
- Recommender systems

Unsupervised

- Clustering
- Association analysis
- LDA

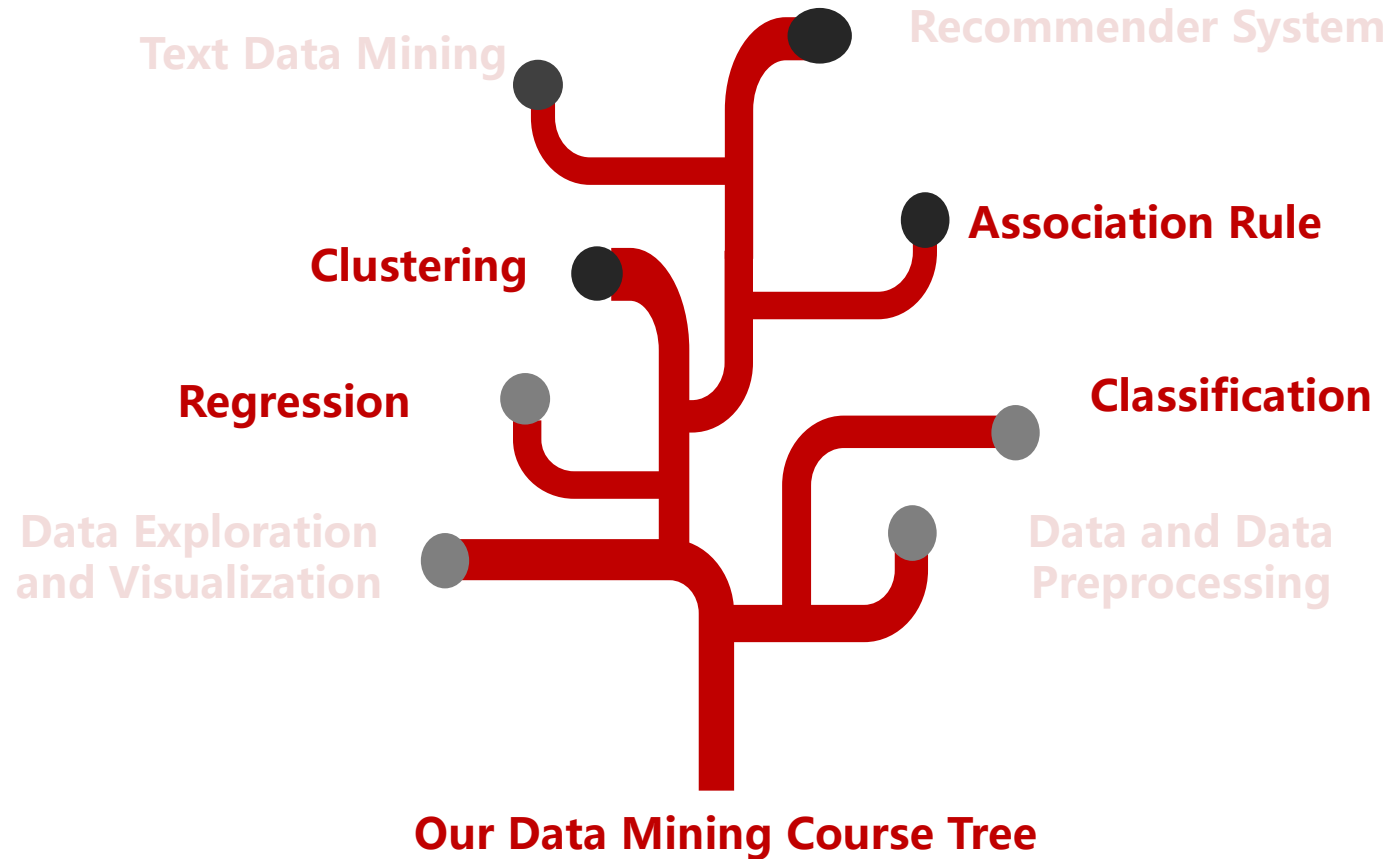
Topics Covered



Our Data Mining Course Tree

- Data Preprocessing
 - Attribute transformation
 - Feature creation
 - Feature selection
 - Dimensionality reduction
- Visualization

Topics Covered



Classification

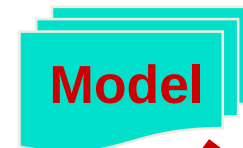
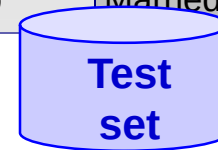
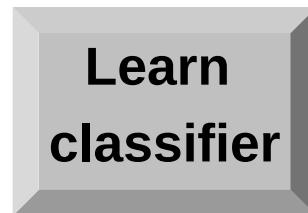
Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

categorical

categorical

continuous

class



Predicted classes

Refund	Marital Status	Taxable Income	Cheat
No	Single	75K	?
Yes	Married	50K	?
No	Married	150K	?
Yes	Divorced	90K	?
No	Single	40K	?
No	Married	80K	?

Classification Problems

- Input: $\mathbf{x} = (x_1, x_2, \dots, x_d)$
- Predict the output (class label) $y \in \mathcal{Y}$
 - Binary classification: $\mathcal{Y} \in \{0, 1\}$
 - Multi-class classification: $\mathcal{Y} = \{1, 2, \dots, C\}$
- **Learn a classification function:** $f(\mathbf{x}) : \mathbb{R}^d \mapsto \mathcal{Y}$
- **Regression:** $f(\mathbf{x}) : \mathbb{R}^d \mapsto \mathbb{R}$

Loss Function

Loss Function:

$\text{loss}(Y, f(X))$ - Measure of closeness between true label Y and prediction $f(X)$

$$\text{loss}(Y, f(X)) = 1_{\{f(X) \neq Y\}} \quad \text{0/1 loss}$$

$$L(Y, f(X)) = |Y - f(X)| \quad \text{Absolute Value loss}$$

$$\text{loss}(Y, f(X)) = (Y - f(X))^2 \quad \text{square loss}$$

$$L(Y, P(Y|X)) = -\log P(Y|X) \quad \text{logarithmic loss function}$$

$$\text{loss}_{\text{exp}}[H(x)] = E_{x,y} \left[e^{-yH(x)} \right] \quad \text{exponential loss function}$$

Regression

- Linear Regression
 - OLS/MLE /MAP Estimator
 - Overfitting issue
- Ridge Regression
- Lasso Regression

Least Squares Estimator

$$\hat{\beta} = \arg \min_{\beta} \frac{1}{n} (\mathbf{A}\beta - \mathbf{Y})^T (\mathbf{A}\beta - \mathbf{Y}) = \arg \min_{\beta} J(\beta)$$

$$J(\beta) = (\mathbf{A}\beta - \mathbf{Y})^T (\mathbf{A}\beta - \mathbf{Y})$$

$$\left. \frac{\partial J(\beta)}{\partial \beta} \right|_{\hat{\beta}} = 0$$

$$\Rightarrow (\mathbf{A}^T \mathbf{A}) \hat{\beta} = \mathbf{A}^T \mathbf{Y}$$

$$\frac{\partial \beta^T \mathbf{x}}{\partial \mathbf{x}} = \beta$$

$$\frac{\partial \mathbf{x}^T \mathbf{x}}{\partial \mathbf{x}} = 2\mathbf{x}$$

$$\frac{\partial \mathbf{x}^T \mathbf{A} \mathbf{x}}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^T) \mathbf{x}$$

MLE: Gaussian Distribution

$$Y \sim \mathcal{N}(\mathbf{A}\beta, \sigma^2 \mathbf{I}) \quad \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}$$



$$p(Y|\mathbf{A}\beta, \sigma^2 \mathbf{I}) = \frac{1}{(\sqrt{2\pi}\sigma)^n} \exp \left\{ -\frac{1}{2\sigma^2} (Y - \mathbf{A}\beta)^T (Y - \mathbf{A}\beta) \right\}$$



$$\log p(Y|\mathbf{A}\beta, \sigma^2 \mathbf{I}) = -\frac{1}{2\sigma^2} (Y - \mathbf{A}\beta)^T (Y - \mathbf{A}\beta) - \text{const}$$

Linear Regression: MAP

- In least square estimator, we have

$$\hat{\beta} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y}$$

- What if $(\mathbf{A}^T \mathbf{A})$ is not invertible ?
- We are going to try MAP estimator

$$\hat{\beta}_{\text{MAP}} = \arg \max_{\beta} \underbrace{\log p(\{(X^{(i)}, Y^{(i)})\}_{i=1}^n | \beta, \sigma^2)}_{\text{log likelihood}} + \underbrace{\log p(\beta)}_{\text{log prior}}$$

Feature Selection and Regression

- R^2
- Ridge/Lasso Regression

$$\hat{\beta}_{\text{MAP}} = \arg \min_{\beta} \sum_{i=1}^n (Y^{(i)} - X^{(i)}\beta)^2 + \lambda \|\beta\|_2^2 \quad \text{Ridge Regression}$$

$\downarrow$
constant(σ^2, τ^2)

$$\hat{\beta}_{\text{MAP}} = \arg \min_{\beta} \sum_{i=1}^n (Y^{(i)} - X^{(i)}\beta)^2 + \lambda \|\beta\|_1$$

$\downarrow$
constant(σ^2, t)

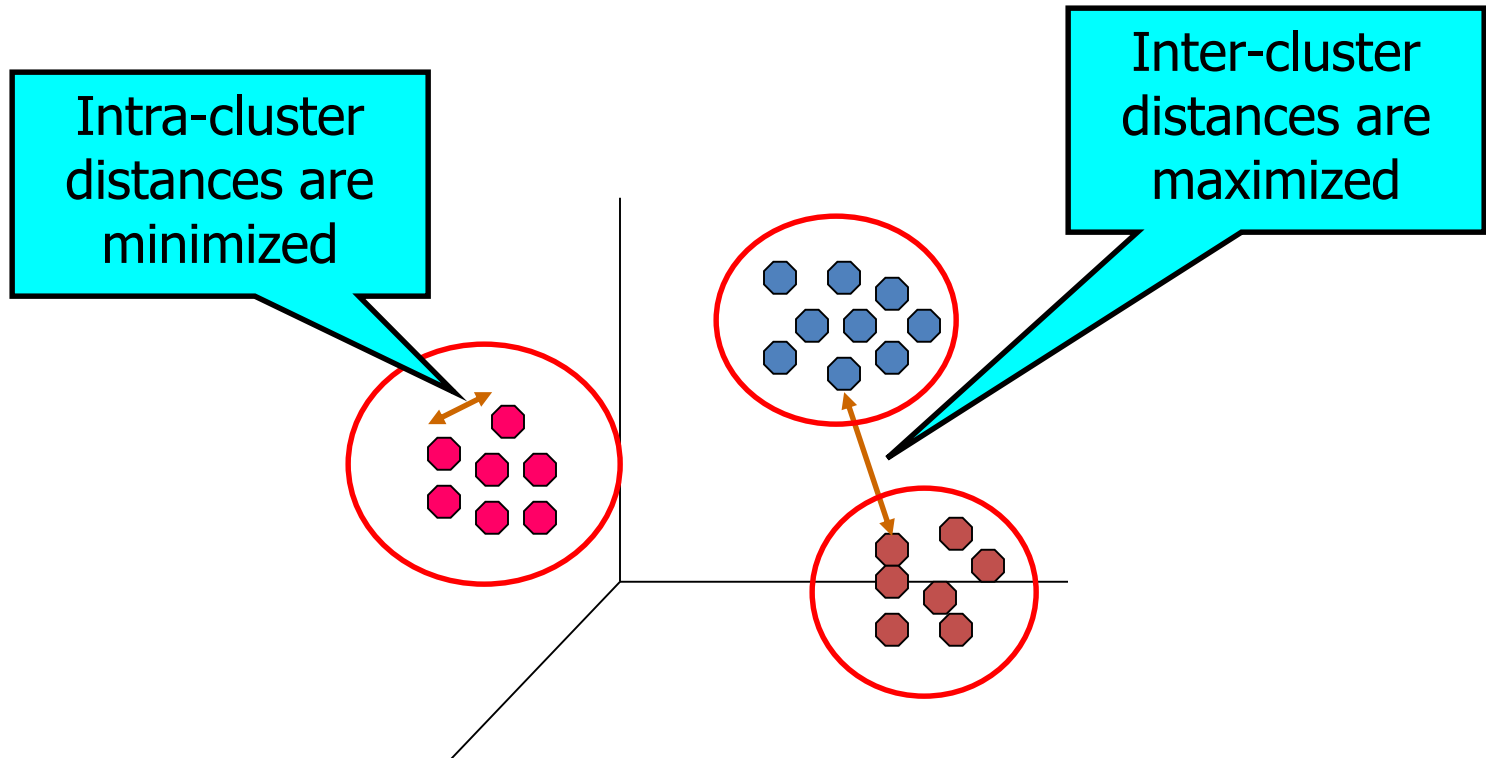
Lasso Regression

Classification

- KNN
- Naïve Bayes
- Logistic Regression
- SVM
 - SVM classifiers for linearly separable classes
 - SVM classifiers for non-linearly separable classes
 - SVM classifiers for nonlinear decision boundaries
- Decision Tree
 - ID3, C4.5, CART
 - Entropy/Gini Index/Maximum Error
 - Pruning and overfitting
- Radom Forest (Boosting)
- Adaboost
- Classifiers/Model Evaluation

What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



Types of clustering

- Partitional
 - Data points divided into finite number of *partitions* (non-overlapping subsets)
- Hierarchical
 - Data points arranged in tree structure that expresses a continuum of similarities and clustering

Clustering

- Similarity and density measurement
- K-means and its variant model
- Hierarchical Clustering
- DBSACN
- Cluster Validation

Association Rule

- Given:
 - set of records each of which contain some number of items from a given collection
- Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

{Milk} --> {Coke}

{Diaper, Milk} --> {Beer}

Association Rule

- Basic Concepts

- Item, Confidence, Support

- Apriori Algorithm

- Frequent Item generation, rule generation

- Mining with Multiple Minimum Supports

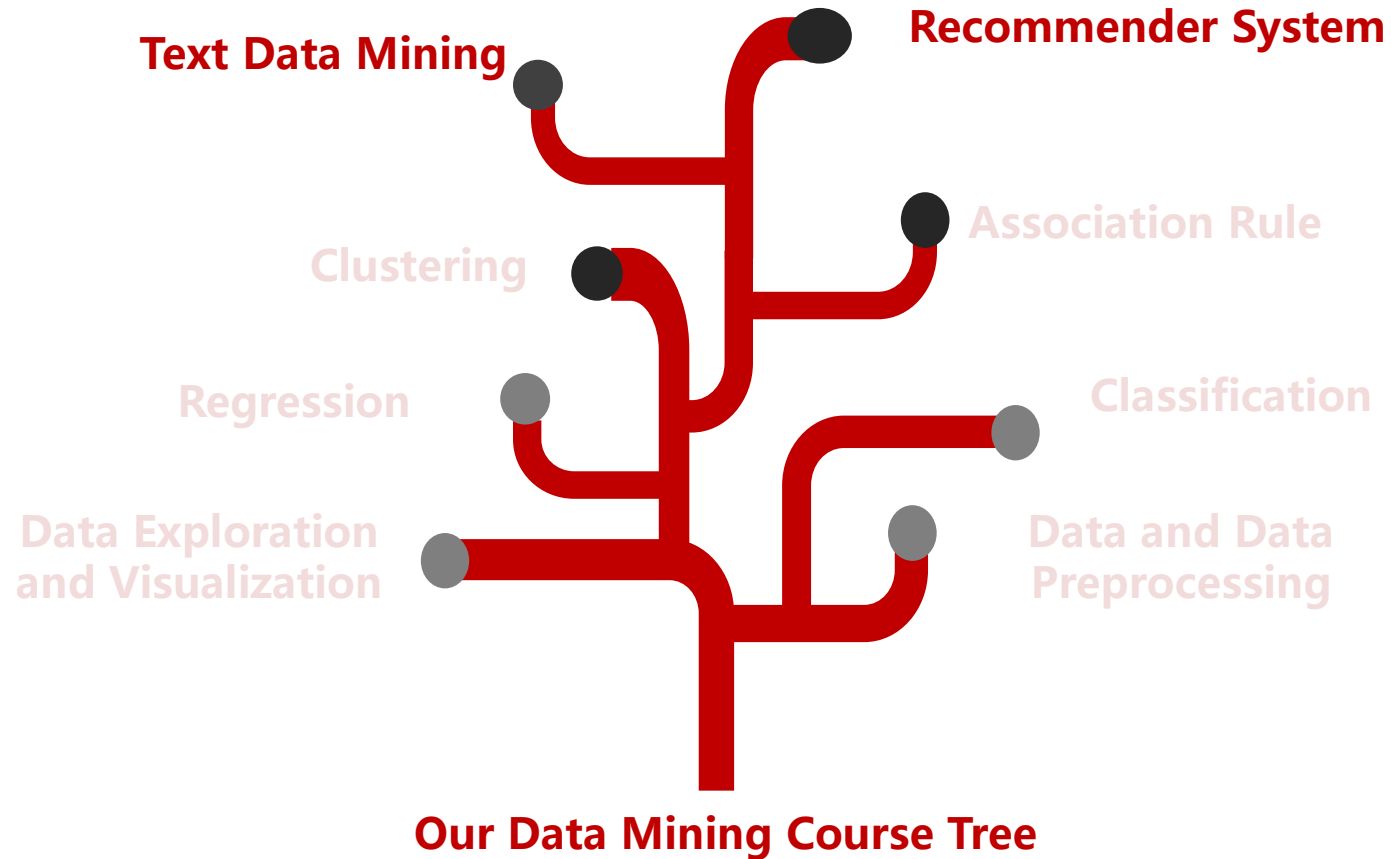
- Mining Class Association Rules

- FP-Growth Method

- Maximal and Closed Frequent Item set

- Sequential Pattern Mining

Topics Covered



Recommender System

Domain: Some field of activity where **users** buy, view, consume, or otherwise experience **items**

Process:

1. users provide ratings on items they have experienced
2. Take all $\langle \text{user}, \text{item}, \text{rating} \rangle$ data and build a predictive model
3. For a user who hasn't experienced a particular item, use model to predict how well they will like it (i.e. predict rating)

Recommender System

- Content-based method
- Collaborative filtering method
 - User based nearest neighbor
 - Item based nearest neighbor
 - Latent factor model

Text Mining

- Text data mining process
- TF-IDF
- LDA
- Classification/cluster/association analysis
- Text mining in social media data

The End

■ THANK YOU

Any questions?

